

How does big data system works

Task 1: "How Do Big Systems Store Data?"

Challenge:

- Search: Do apps like Instagram, Uber, or Amazon use **SQL databases** or something else?
- Summarize the kind of **database structure** big companies use and why.
- Try to find one that uses **SQL Server** specifically.

The company	The data base language
Instagram	PostgreSQL (SQL) Cassandra (NoSQL)
Uber	MySQL (SQL) MongoDB Redis
Amazon	DynamoDB (NoSQL) Aurora

What kind of database structure do big companies use and why?

Large companies typically use a **hybrid database structure**, combining **SQL** and **NoSQL** databases. SQL databases are ideal for handling **structured and relational data**—like user profiles, payments, and order histories—because they provide strong consistency and reliability. On the other hand, NoSQL databases are used for data that is **high in volume, unstructured, or rapidly changing**, such as logs, messages, or real-time analytics, as they offer **greater speed and scalability**.

Example of a company that uses SQL Server specifically

One well-known example of a company using **SQL Server** is **Microsoft** itself. They use SQL Server in many of their enterprise services, especially within **Azure SQL Database**, which is a cloud-based relational database service. Beyond Microsoft, many **government agencies, banks, and large enterprises** rely on SQL Server because of its **security, robust transaction handling, and integration with other Microsoft tools**.

Task 2: "Do Video Games Use Databases?"

Challenge:

- How do online games (like PUBG, Fortnite, etc.) store **player progress, points, and profiles**?
- Do they use SQL databases or something else?
- Find an example of a game that uses a **relational database**.

Search hints:

- "How do video games store player data?"
- "Relational databases in gaming industry"

Do Video Games Use SQL or Other Databases

Modern video games often use a combination of database types. SQL databases are used to manage structured and relational data such as player accounts, achievements, and rankings. On the other hand, NoSQL databases are preferred for handling massive, fast-changing, or unstructured data like real-time game events or chat messages. Most large-scale games use both types depending on the use case.

Example of a Game Using a Relational Database

A good example of a game that uses a relational database is **League of Legends**. It utilizes **MySQL** to manage detailed information about player accounts, match records, and performance statistics. This relational structure helps the developers query complex data efficiently and maintain high data integrity for competitive gameplay.

Task 3: "Can a Database Work Without the Internet?"

Challenge:

- Search how databases are used in **offline systems**, like in **military bases** or **airplanes**.
- What kind of database do they use when no internet is available?
- Can SQL Server be used offline?

Search hints:

- "Offline SQL database use cases"
- "Can SQL Server run without internet?"

How are databases used in offline systems like military bases or airplanes?

In offline environments such as military bases, submarines, or airplanes, databases are often deployed locally on secure, isolated servers. These databases handle critical tasks like tracking inventory, managing logs, and recording operational data without relying on an internet connection. Data is stored locally and synced with central systems later when internet access becomes available.

What kind of databases are used when no internet is available?

When there is no internet connection, organizations rely on **on-premises databases** such as **SQLite**, **Microsoft SQL Server**, or **Oracle Database** installed directly on physical machines or local networks. These databases are fully functional offline and are designed to handle read/write operations without needing cloud services. The key is that the database engine must be installed locally and the system must be self-contained.

Can SQL Server be used offline?

Yes, **SQL Server can work completely offline**. It is a standalone database engine that runs on local servers or machines without requiring internet access. Many businesses and government agencies use SQL Server in secure, isolated environments for performance, data control, and security reasons. It only requires internet for specific tasks like remote backups, cloud integration, or updates—but basic database operations function perfectly offline.

Task 4: "What Makes Online Stores So Fast?"

Challenge:

- How do e-commerce platforms (like **Amazon** or **Noon**) organize their product and order data for fast searching?
- Do they use **relational databases**, **cloud**, or both?
- What challenges do they face in storing millions of products?

Search hints:

- "How does Amazon store product data?"
- "SQL vs NoSQL for e-commerce websites"
- "Cloud database for online shopping platforms"

How do platforms like Amazon or Noon organize their data for fast searching?

E-commerce platforms like **Amazon** and **Noon** organize their product and order data using a combination of **indexing, caching, and database partitioning**. They structure the data to allow quick lookup using advanced search algorithms. Product metadata, user preferences, and stock levels are stored in databases that are optimized for speed, and frequently accessed data is cached in memory for ultra-fast responses.

Do they use relational databases, cloud, or both?

These platforms usually rely on **both relational and NoSQL databases**, deployed in **cloud environments** for flexibility and scalability. **Relational databases** (like MySQL or PostgreSQL) are used for structured data such as orders and payments. **NoSQL databases** (like DynamoDB or Cassandra) manage large-scale unstructured data like customer behavior, product reviews, and inventory tracking. The **cloud** plays a crucial role by offering scalable resources that automatically adjust to traffic loads.

What challenges do they face with storing millions of products?

One of the biggest challenges is handling massive amounts of data efficiently without slowing down search or checkout experiences. These platforms must ensure real-time updates, data consistency, and minimal downtime while supporting millions of users simultaneously. They also face issues like data duplication, maintaining search relevance, and ensuring secure, scalable infrastructure capable of handling seasonal traffic spikes like Black Friday.

Task 5: "How Do Banks Protect Their Data?"

Challenge:

- Search how **banking systems** use databases to protect sensitive customer data.
- What database features help with **security and integrity**?
- Find one bank or financial system that uses **Oracle** or **SQL Server**.

How Do Banks Protect Their Data?

How do banking systems use databases to protect sensitive customer data?

Banking systems rely heavily on secure and robust databases to protect customer data, such as account numbers, balances, and transaction history. These databases are hosted on highly secured servers, often within private networks or data centers with strong physical and digital protections. Data is encrypted both at rest and in transit, and access is tightly controlled using multi-layer authentication systems to prevent unauthorized access.

What database features help with security and integrity?

Databases used in banking systems offer built-in features that ensure security and integrity, including role-based access control (RBAC), data encryption, audit logging, transaction rollbacks, and referential integrity constraints. These features ensure that only authorized personnel can access or modify data, and any changes are tracked and reversible. Features like ACID compliance (Atomicity, Consistency, Isolation, Durability) ensure that financial transactions are processed correctly and reliably.

Example of a bank using Oracle or SQL Server

Many global banks use Oracle Database due to its advanced security features and scalability. For example, Bank of America uses Oracle to manage large-scale data securely. Additionally, SQL Server is widely used in regional banks and financial institutions, especially those that operate in Microsoft-based environments, such as Emirates NBD in the Middle East, which uses Microsoft technologies including SQL Server for part of its operations.

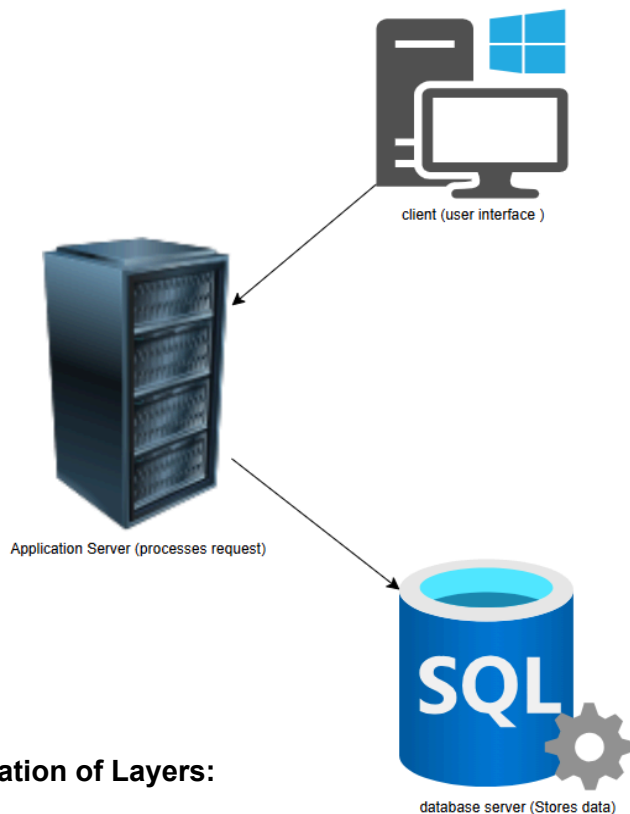
Task 6: "Draw Me the System!"

Challenge:

- Based on what you know about **clients**, **application server**, and **DB server**, search for and draw a diagram that shows the **three-tier architecture**.
- Add a short explanation: what each layer does.

Search hints:

- "Three-tier architecture database diagram"
- "Client application database interaction example"



Explanation of Layers:

1. Client (User Interface):

This is what the user interacts with—such as a web browser or mobile app. It sends user requests to the application server and displays the results.

2. Application Server (Logic Layer):

This layer processes the requests coming from the client, applies business logic, and communicates with the database server when needed.

3. Database Server (Data Layer):

This layer stores and manages the actual data. It responds to queries from the application server, such as fetching user records or saving updates.

